

Protein function classification via support vector machine approach

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Abstract

Support vector machine (SVM) is introduced as a method for the classification of proteins into functionally distinguished classes. Studies are conducted on a number of protein classes including RNA-binding proteins; protein homodimers, proteins responsible for drug absorption, proteins involved in drug distribution and excretion, and drug metabolizing enzymes. Testing accuracy for the classification of these protein classes is found to be in the range of 84–96%. This suggests the usefulness of SVM in the classification of protein functional classes and its potential application in protein function prediction.

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1. Introduction

Knowledge about the function of proteins is essential in the understanding of biological processes [1–4]. With the accumulation of sequence information, attention has been paid to the development of methods for the prediction of protein function [5,6] from sequence. These methods determine the function of a protein by classifying it into a specific protein family or functional class based on direct sequence similarity analysis or the search of conserved sequence motifs. In the absence of clear sequence or structural similarities, the criteria for inclusion of distantly

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related proteins into a protein functional class becomes increasingly arbitrary. Therefore, alternative classification methods may need to be explored in facilitating the study of protein function.

Recently a relatively new classification method, support vector machines (SVM) [7,8], has been used for the prediction of protein–protein interaction [9], protein fold recognition [10,11], and protein structure prediction [12–14]. Instead of directly analyzing sequences, SVM classification method used in these studies is based on the analysis of physicochemical properties of a protein generated from its sequence. Such an approach may be extended to the classification of proteins into functional classes, and thus facilitating the prediction of function of these proteins. Proteins of specific functional class share common structural features essential for performing similar functions such as transport, metabolism, and binding induced signal transduction. The same structural features responsible for protein folding are thus expected to be part of the elements for determination of protein functions. Therefore, as a method successfully used in protein fold recognition, SVM may be potentially used for protein function classification.

Some of function-related features might be captured by the residue properties such as hydrophobicity, polarity, polarizability, charge, surface tension, secondary structure, normalized van der Waals volume and solvent accessibility which have been used in the construction of the SVM feature vectors for proteins [9,10]. Given sufficient samples of proteins of a functional class, SVM can then be trained and used for the classification of new proteins against that class. This approach is expected to be capable of classification of distantly related proteins as well as other proteins to a certain extent. Therefore SVM may be explored as a general method for protein function prediction by means of classification of proteins into specific functional classes.

In this work, the usefulness of SVM for protein function prediction is extensively tested by applying it to the classification of a variety of functionally distinguished classes of proteins. These include RNA-binding proteins, protein homodimers, proteins responsible for drug absorption, proteins involved in drug distribution, and drug metabolizing enzymes. These classes of proteins are suitable for testing SVM as they represent proteins of diversely different functions that cover protein synthesis regulation, protein self-association, drug transport and drug metabolism.

2. Support vector machine

In addition to classification of proteins, SVMs have been employed in a wide range of real-world problems such as text categorization [15–18], hand-written digit recognition [7,19], tone recognition [20–22], image classification and object detection [23–28], sub-storm and flood stage forecasting [29,30], cancer diagnosis [31–34], glaucoma diagnosis [35], microarray gene expression data analysis [36–38]. It has been shown that SVM is consistently superior to other supervised learning methods [38–40].

When used for classification, SVMs separate a given known set of $\{+1, -1\}$ labeled training data via a hyperplane that is maximally distant from the positive samples and negative samples (known as Optimal Separating Hyperplane, OSH, see Fig. 1), then ‘plot’ the test data at the high dimensional space, distinguish whether it belongs to positive or negative according to the OSH. For most of real-world problems, seem not to be linearly separable, SVMs can work in combination with the technique of ‘kernels’, that automatically realizes a non-linear mapping onto a

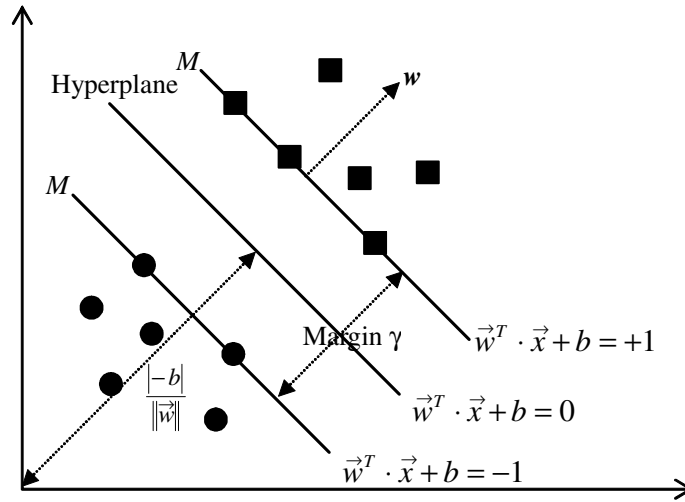


Fig. 1. Definition of hyperplane and margin. The circular dots and square dots represent samples of class -1 and class $+1$, respectively.

feature space. The OSH found by the SVM in feature space corresponds to a non-linear decision boundary in the input space.

2.1. Linear classification

Let the training data of two separable classes with n samples be represented by $(\vec{x}_1, y_1), (\vec{x}_2, y_2), \dots, (\vec{x}_n, y_n)$, $i = 1, 2, \dots, n$, where $\vec{x}_i \in R^N$ is an N dimensional space, and $y_i \in \{-1, +1\}$ is the class index. Given a weight vector $\vec{w}$ and bias b (Fig. 1), the two classes can be separated by two margins parallel to the hyperplane

$$\vec{w}^T \cdot \vec{x}_i + b \geq 1, \quad y_i = +1, \quad (1)$$

$$\vec{w}^T \cdot \vec{x}_i + b \leq -1, \quad y_i = -1, \quad (2)$$

where $\vec{w} = (w_1, w_2, \dots, w_n)^T$ is a vector of n elements. Inequalities (1), (2) can be combined into a single inequality

$$y_i(\vec{w}^T \cdot \vec{x}_i + b) \geq 1, \quad i = 1, 2, \dots, n. \quad (3)$$

There exists a number of hyperplanes for each group of training data. The classification objective of SVM is to determine an optimal weight $\vec{w}_0$ and an optimal bias b_0 such that the selected hyperplane separates the training data with maximum margin. The hyperplane determined by $\vec{w}_0$ and b_0 is called OSH.

The equation of any given hyperplane can be written as

$$\vec{w}^T \cdot \vec{x}_i + b = 0 \quad (4)$$

and the distance between the two corresponding margins is

$$\gamma(\vec{w}, b) = \min_{\{\vec{x}|y=+1\}} \frac{\vec{x}^T \cdot \vec{w}}{\|\vec{w}\|} - \max_{\{\vec{x}|y=-1\}} \frac{\vec{x}^T \cdot \vec{w}}{\|\vec{w}\|}. \quad (5)$$

The OSH can be found by maximizing the above distance or by minimizing the norm of $\|\vec{w}\|$ under inequality constraints (3), and

$$\gamma_{\max} = \gamma(\vec{w}_0, b_0) = \frac{2}{\|\vec{w}_0\|}. \quad (6)$$

The saddle point of the following Lagrangean gives solutions to the above optimization problem

$$L(\vec{w}, b, \alpha) = \frac{1}{2} \vec{w}^T \cdot \vec{w} - \sum_{i=1}^n \alpha_i [y_i (\vec{w}^T \cdot \vec{x}_i + b) - 1], \quad (7)$$

where $\alpha_i \geq 0$ are Lagrange multipliers. The solution of this optimization Quadratic Programming (QP) problem requires that the gradient of $L(\vec{w}, b, \alpha)$ with respect to $\vec{w}$ and b vanishes, which gives the following conditions by the calculation of $\frac{\partial L}{\partial \vec{w}}|_{\vec{w}=\vec{w}_0} = 0$ and $\frac{\partial L}{\partial b}|_{b=b_0} = 0$:

$$\vec{w}_0 = \sum_{i=1}^n \alpha_i y_i \vec{x}_i, \quad (8)$$

$$\sum_{i=1}^n \alpha_i y_i = 0. \quad (9)$$

By substituting Eqs. (8) and (9) into Eq. (7), the QP problem becomes maximization of the following expression:

$$L(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\vec{x}_i^T \cdot \vec{x}_j) \quad (10)$$

under the constraints $\sum_{i=1}^n \alpha_i y_i = 0$ and $\alpha_i \geq 0, i = 1, 2, \dots, n$.

The points situated at the two optimal margins have non-zero coefficients α_i among the solutions to Eq. (10), and are called support vectors (SV). The bias b_0 can be calculated as follows:

$$b_0 = -\frac{1}{2} \left(\min_{\{\vec{x}_i | y_i = +1\}} \vec{w}_0^T \cdot \vec{x}_i + \max_{\{\vec{x}_i | y_i = -1\}} \vec{w}_0^T \cdot \vec{x}_i \right). \quad (11)$$

After determination of SV and bias, the decision function that separates the two classes can be written as

$$f(\vec{x}) = \text{sign} \left[\sum_{i=1}^n \alpha_i y_i \vec{x}_i^T \cdot \vec{x} + b_0 \right] = \text{sign} \left[\sum_{\text{SV}} \alpha_i y_i \vec{x}_i^T \cdot \vec{x} + b_0 \right]. \quad (12)$$

2.2. Non-linear classification

As real-world problems are usually non-linear, the following approach has been introduced into SVM to deal with these problems. The original training data $\vec{x}$ in the input space $\mathbf{X}$ is projected into a high-dimensional feature space $\mathbf{F}$ via a Mercer kernel operator K [7], and the OSH is constructed in this feature space. In mathematical terms, the set of classifiers is transformed into the form

$$f(\vec{x}) = \text{sign} \left[\sum_{i \in \{\text{SV}\}} \alpha_i y_i K(\vec{x}_i, \vec{x}) + b_0 \right], \quad (13)$$

where K is a symmetric positive definite function, which satisfies Mercer's conditions

$$K(\vec{x}, \vec{y}) = \sum_{m=1}^{\infty} \alpha_m \phi(\vec{x})^T \cdot \phi(\vec{y}), \quad \alpha_m \geq 0, \quad (14)$$

$$\int \int K(\vec{x}, \vec{y}) g(\vec{x}) g(\vec{y}) d\vec{x} d\vec{y} > 0, \quad \int g^2(\vec{x}) d\vec{x} < \infty$$

the Kernel represents a legitimate inner product in input space

$$K(\vec{x}, \vec{y}) = \phi(\vec{x})^T \cdot \phi(\vec{y}). \quad (15)$$

In the $\mathbf{F}$ -space, the dual Lagrangian, given in Eq. (10), is

$$L(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j K(\vec{x}_i, \vec{x}_j) - \lambda \sum_{i=1}^n \alpha_i y_i \quad (16)$$

$$\text{subject to } \sum_{i=1}^n \alpha_i y_i = 0 \quad \text{and} \quad \alpha_i \geq 0, \quad i = 1, 2, \dots, n$$

and the decision function is

$$f(\vec{x}) = \text{sign} \left[\sum_{i \in \{\text{SV}\}} \alpha_i y_i K(\vec{x}_i, \vec{x}) + b_0 \right], \quad (17)$$

where

$$b_0 = -\frac{1}{2} \left\{ \min_{\{\vec{x}_i | y_i = +1\}} \left(\sum_{j \in \{\text{SV}\}} \alpha_j y_j K(\vec{x}_i, \vec{x}_j) \right) + \max_{\{\vec{x}_i | y_i = -1\}} \left(\sum_{j \in \{\text{SV}\}} \alpha_j y_j K(\vec{x}_i, \vec{x}_j) \right) \right\}. \quad (18)$$

A number of candidate kernel functions have been used in SVM, including polynomial $K(\vec{x}, \vec{y}) = (1 + \vec{x} \cdot \vec{y})^d$, Gaussian RBF $K(\vec{x}, \vec{y}) = \exp\left(-\frac{\|\vec{x} - \vec{y}\|^2}{2\sigma^2}\right)$, exponential RBF $K(\vec{x}, \vec{y}) = \exp\left(-\frac{\|\vec{x} - \vec{y}\|}{2\sigma^2}\right)$ and their combinations by summing kernels or tensor products of kernels [41]. In this work, Gaussian kernel function is used.

3. Feature vectors and accuracy measure

3.1. Feature vectors

Construction of feature vectors is the key to successful SVM classification. For each protein sequence, feature vectors are assembled from encoded representations of tabulated residue properties including amino acids composition, hydrophobicity, normalized van der Waals

Table 1
Physicochemical feature vectors and their dimensionality

Parameter name	Dimension
Amino acids composition	20
Hydrophobicity	21
Normalized van der Waals volume	21
Polarity	21
Polarizability	21
Charge	21
Surface tension	21
Secondary structure	21
Solvent accessibility	21

These vectors are generated from protein sequence.

volume, polarity, polarizability, charge, surface tension, secondary structure and solvent accessibility for each residue in sequence. The use of some of the features is motivated by previous studies of proteins [9,10]. The composition and dimensionality of feature vectors are shown in Table 1.

3.2. Accuracy measure

Accuracy of results from discriminative methods is commonly measured by the quantity of True Positives (TP), True Negatives (TN), False Positives (FP), False Negatives (FN) [10,42]. In addition to these quantities, standard sensitivity, specificity and overall accuracy (Q) performance measures defined by

$$\text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN}), \quad (19)$$

$$\text{Specificity} = \text{TN}/(\text{TN} + \text{FP}), \quad (20)$$

$$Q = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (21)$$

are also useful in assessing the prediction accuracy [43]. All these quantities are used in the evaluation of SVM classification of proteins in this work.

4. Protein sample selection

Seven functional classes of proteins are used for testing the capability of our SVM program [44] in the classification of proteins into functional classes. These are RNA-binding proteins, protein homodimers, drug absorption proteins, drug delivery proteins, drug excretion proteins, class-I drug metabolizing enzymes, and class-II drug metabolizing enzymes. All distinct members in each family are used to construct positive samples of the training, testing and independent evaluation sets for each family. The negative samples for the training, testing and independent evaluation sets

of each family are selected from seed proteins of the curated protein families in the Pfam database (<http://www.sanger.ac.uk/Software/Pfam/>) excluding those that belong to the family under study.

Training sets of both positive and negative samples are further screened so that only essential proteins that optimally represent each family are retained. The SVM training system for each family is optimized and tested by using separate testing sets of both positive and negative samples composed of all the remaining distinct proteins of a family and those outside the family, respectively. The performance of SVM classification is further evaluated by using independent sets of both positive and negative samples composed of all the remaining proteins of a family and those outside the family, respectively. No duplicate protein is used in the training, testing and independent evaluation set for each family.

5. Results and discussion

5.1. RNA-binding proteins

RNA-binding proteins regulate protein synthesis by interacting with various RNAs at different stages of protein synthesis. All RNA-binding proteins used in this experiment are obtained from protein sequence database Swissprot (<http://www.expasy.ch/sprot>). This database currently contains 2623 entries of proteins known to bind to RNA, of which 2223 entries are found to have no unknown amino acid marked by 'X'. There are 742 distinct proteins which are used as positive samples for the training set. In the remaining entries, there are 741 distinct proteins which are used as positive samples in the test set. Upon subtracting positive samples for the training and testing sets, there are 740 proteins left which are used as independent evaluation set. After optimization of SVM training system, the number of proteins in the training and testing sets become 871 and 612, respectively. The number of proteins in the negative samples of training, testing and independent evaluation sets is 1120, 1157, and 978, respectively.

5.2. Homodimers

Certain proteins form homodimers (self-association) or heterodimers (cross-association) when activated which enables the newly formed dimer to interact with other proteins. Homodimers used in this work are from Protein Data Bank (PDB) (<http://www.rcsb.org/pdb/>) and Swiss-prot. Duplications are removed and only those entries which explicitly specified as dimer in Swiss-prot database are used as positive samples. Additional dimers described in the literature [45] are also added. The total number of positive samples is 529 homodimers. There are 219 distinct proteins which are used as positive samples for the training set. In the remaining entries, there are 218 distinct proteins which are used as positive samples in the test set. Upon subtracting positive samples for the training and testing sets, there are 92 proteins left which are used as independent evaluation set. After optimization of SVM training system, the number of proteins in the training and testing sets become 349 and 88, respectively. The number of proteins in the negative samples of training, testing and independent evaluation sets is 368, 97, and 91, respectively.

5.3. *Proteins responsible for drug absorption, distribution, metabolism and excretion*

Drug absorption, distribution, metabolism and excretion (ADME) often involve the interaction of a drug or its metabolites with specific proteins. The ADME associated proteins used in this work are selected from the drug ADME-associated protein database [46]. All the corresponding entries of these proteins are searched from Swissprot. Five classes of ADME associated proteins are studied in this work. The first class includes drug absorption proteins, which has 405 entries. The second class is composed of proteins responsible for drug delivery into the site of action, which has 2870 entries. The third class consists of proteins involved in drug transport back into the blood stream or excretion out of body, which has 428 entries. The fourth class contains class-I drug metabolizing enzymes, which has 298 entries. These enzymes are responsible for the oxidation, reduction and hydrolysis of drugs. The fifth class is a collection of class-II drug metabolizing enzymes, which has 41 entries. These enzymes are involved in the conjugation of drugs which become inactive and are subsequently removed from the body.

After optimization of SVM training system, the number of positive samples in the training and testing sets are 228 and 88 for drug absorption, 916 and 404 for drug delivery, 262 and 70 for drug excretion, 117 and 97 for class-I drug metabolism, and 84 and 63 for class-II drug metabolism, respectively. The number of positive samples for the independent evaluation sets is 124, 629, 150, 84 and 49 for the drug absorption, drug delivery, drug excretion, class-I drug metabolism and class-II drug metabolism protein family, respectively. The number of proteins in the negative samples of training, testing and independent evaluation sets is 1651, 2222, and 1436 for drug absorption, 2037, 1843, and 1441 for drug delivery, 2631, 2902, and 2263 for drug excretion, 3032, 2607, and 2313 for class-I drug metabolism and 3031, 2604, and 2322 for class-II drug metabolism, respectively.

5.4. *SVM classification performance analysis*

The results for the classification of each of these protein classes are given in Table 2. In Table 2, P and N represent positive and negative samples. The overall accuracy of protein classification of the respective independent evaluation sets range from 86.5% to 99.4%. A direct comparison with results from previous studies is inappropriate because of differences in the specific aspects of proteins, use of dataset, descriptors and classification methods. Nonetheless, a tentative comparison may provide some crude estimate regarding the level of accuracy of our method with respect to those achieved by other studies. The range of accuracy from our study is comparable to other SVM studies of proteins [9,12,13] and other systems [20,24,29,34,39]. Moreover, the sensitivity and specificity are the range of 44.0–96.7% and 90.1–99.6%, respectively. Therefore our study indicates the capability of SVM in classification of proteins into specific functional classes.

In this work, a total of nine feature vectors are used to describe each protein, as motivated by previous studies of proteins [9,10]. It has been reported that, not all feature vectors play equal role, some seem to play more prominent role in specific aspects. For instance, in protein fold recognition, composition, secondary structure and hydrophobicity have been found to play more prominent role than other feature vectors [10]. It is of interest to examine which feature vectors play more prominent role in protein function classification. Our analysis indicates that the

Table 2
Test results for different protein functional classes

Protein class	Training set		Testing set				Independent evaluation set						
	Posi- tive	Nega- tive	Positive		Negative		Positive		Negative		Sensitivity (%)	Specificity (%)	Q (%)
			TP	FN	TN	FP	TP	FN	TN	FP			
RNA-binding protein	871	1120	610	2	1153	4	613	127	898	80	82.8	91.8	88.0
Homodimer	349	368	88	0	97	0	88	3	82	9	96.7	90.1	93.4
Drug absorption	228	1651	53	0	2219	3	83	41	1377	59	66.9	95.8	93.6
Drug delivery	916	2037	404	5	1827	16	431	198	1360	81	68.5	94.3	86.5
Drug excretion	262	2631	59	11	2900	2	66	84	2102	161	44.0	92.8	89.8
Class-I drug metabolism	117	3032	82	15	2606	1	76	8	2190	123	90.4	94.6	94.5
Class-II drug metabolism	84	3031	53	10	2600	4	42	7	2315	7	85.7	99.6	99.4

composition, charge, polarity, hydrophobicity play more prominent role than other feature vectors. Charge and polarity are important for electrostatic interactions and hydrogen-bonding, which are two important factors for the binding of a protein with its partner. Therefore their importance in protein function is expected, even though they do not play prominent role in fold recognition. Unlike in the case of protein folding recognition, secondary structure does not appear to be a prominent feature vector. Secondary structure is the basis for structural frame of protein fold. It is however not necessarily the essential element for the binding of a partner to a protein. Therefore, it is understandable that this feature vector is not a prominent one.

Several factors may affect the prediction accuracy. One is the diversity of protein samples. It is likely that not all possible types of proteins are adequately represented in the training sets for some or all of the functional classes. This can be improved along with the availability of more protein data. SVM prediction may be further improved by choosing a more refined set of negative samples for each class that include all types of proteins in close proximity the class. Such negative samples enable the generation of truly optimal support vectors. SVM optimization procedure and feature vector selection may also be improved.

6. Conclusion

Testing results on seven protein functional classes suggests that SVM appears to be a potentially useful tool for protein function prediction by means of classification of proteins into specific functional classes. Further works on samples collection for every functional class, refined negative samples selection, and improvement of SVM optimization algorithm and feature vector selection will help in development of SVM into a practical protein function prediction tool.

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